

Missing Data

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Accuracy as a Fraction

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace filtering for Missing values and validation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Missing values and validation. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Validate independently

Write a filter for count values that must be integers from 0 to 10. Process {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

6. Solve the given case

Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 18 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use 'no preference' only when the reader confirms it; otherwise ask or explain uncertainty.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Distinguish zero and unknown

Explain the difference between {"minutes": 0}, {} and {"minutes": None}.

My reasoning and evidence

13. W2 Trace filtering

Predict accepted values and rejections for minutes 4, 0, 61 and "4".

My reasoning and evidence

14. W3 Repair a condition

A program uses if value to accept durations. Give a counterexample and the correct rule.

My reasoning and evidence

15. W4 Validate independently

Write a filter for count values that must be integers from 0 to 10. Process {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

My reasoning and evidence

16. W5 Report the limitation

Only 2 of 5 supplied durations were accepted. Why should the report include this?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Missing Data

A fictional catalogue records counts of available copies under a strict input contract.

Accept count only if `type(value)` is `int` and $0 \leq \text{value} \leq 8$. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

Missing Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve carefully

Use `row.get("count")`; absent field gives None, distinct from 0.



2. Check exact type

Integers 0,4 pass; text "4", Boolean True and None fail.



3. Check range

Integer 9 fails maximum 8.



4. Report validation

Accepted counts [0,4]; four rows rejected with recorded reasons.

CHANGE ONE THING

Replace text "4" with integer 4 through an authorised conversion.

Explain the changed result using the matching lesson.

Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue records counts of available copies under a strict input contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accepted counts [0,4]; four rows rejected with recorded reasons.

1. Report validation
2. Retrieve carefully
3. Check range
4. Check exact type

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report validation Accepted counts [0,4]; four rows rejected with recorded reasons. Testing if value would wrongly discard zero, which is a real valid count. Python's `isinstance(True,int)` is true, so the stated exact-integer contract uses `type(value)` is `int` to reject Boolean input. Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

Practice 2: explain the deciding evidence

Changed-case practice: Replace text "4" with integer 4 through an authorised conversion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Accepted values become [0,4,4]; three rows remain rejected.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Accepted values become [0,4,4]; three rows remain rejected. The converted value now satisfies both type and range checks.

Repair a missing building block

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?. Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf predictions from training examples. Short practice: Choose the best original question and explain why its more balanced split matters here.. Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
